

Unit 1 Practice Test

1

★ Mark for Review

ABC

A line in the xy -plane is defined by the equation $y = ax + b$, where a and b are constants. The line passes through the points $(0, -79)$ and $(5, 91)$. What is the value of $a + b$?

2

★ Mark for Review

ABC

In the xy -plane, the line with equation $ax - \frac{1}{3}y = 8$, where a is a constant, passes through the point $(2, 6)$. What is the x -coordinate of the x -intercept of the line?

3

★ Mark for Review

ABC

The table shows three values of x and their corresponding values of y , where p is a constant. There is a linear relationship between x and y . What is the slope of the line that represents this relationship in the xy -plane?

x	y
-6	$-114 - p$
-2	$-57 - p$
2	$-p$

A. $\frac{57}{4}$

B. $\frac{57}{4}$

C. $\frac{p - 57}{4}$

D. $\frac{2p + 57}{4}$

4

★ Mark for Review

ABC

The graph of the equation $5x - 8y = k$, where k is a constant, is a line in the xy -plane. The line has an x -intercept at $(m, 0)$ and a y -intercept at $(0, n)$, where m and n are constants. What is the value of $\frac{m}{n}$?

A. $-\frac{8}{5}$

B. $\frac{5}{7}$

C. $\frac{5}{8}$

D. $\frac{8}{6}$

5

★ Mark for Review

ABC

In the xy -plane, the graph of the linear function f has a negative slope and a y -intercept at $(0, k)$, where k is a positive constant. Which of the following must be true?

- I. If $a > b$, then $f(a) < f(b)$
- II. If $a < 0$, then $f(a) > k$
- III. If $a > 0$, then $f(a) > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II, and III

6

★ Mark for Review

ABC

In the xy -plane, the points $(-3, 5)$ and $(6, 8)$ lie on line ℓ . Which of the following points also lies on line ℓ ?

A. $(27, 15)$ B. $(33, 19)$ C. $(42, 23)$ D. $(54, 27)$ **7**

★ Mark for Review

ABC

During a road trip, Carl used 12 gallons of fuel driving a minivan and 17 gallons of fuel driving a campervan. He traveled a total distance of 590 miles driving both vehicles. The equation $12x + 17y = 590$ represents this situation. Which is the best interpretation of y in this context?

A. The total number of gallons of fuel Carl used driving the campervan

B. The total number of miles Carl drove the campervan for

C. The fuel efficiency, in miles per gallon, of the campervan

D. The average fuel efficiency, in miles per gallon, of the minivan and the campervan

8

★ Mark for Review

ABC

Of the 585 photos in a photo album, 315 are color photos and the rest are black-and-white. The equation $6p + 315 = 585$ describes this situation, where p represents the number of pages that contain black-and-white photos. What is the best interpretation of $6p$ in this context?

A. There are $6p$ black-and-white photos.

B. There are $6p$ photos on each page.

C. There are $6p$ pages in total.

D. There are $6p$ photos in total.

9

★ Mark for Review

ABC

Stacy is playing a game in which she needs to collect a certain number of points to win. The function $p(x) = 52 - 4x$ models the number of additional points Stacy needs to win the game, where x is the number of rounds played. Which of the following is the best interpretation of $p(8) = 20$?

A. Stacy collected a total of 20 points in the first 8 rounds.

B. It will take Stacy 8 rounds to collect the 20 additional points she needs to win the game.

C. After the first 8 rounds, Stacy needs to collect 20 more points to win the game.

D. In the 8th round, Stacy collected 20 more points than she did the previous round.

10

★ Mark for Review

ABC

A party store sells premade banners and custom banners. Yesterday, the store sold a total of 80 banners, earning x dollars in revenue from premade banners and y dollars in revenue from custom banners. The equation $\frac{x}{14} + \frac{y}{25} = 80$ represents this situation. Which of the following is the best interpretation of 14 in this context?

- A. The store sold 14 premade banners yesterday.
- B. The store earns 14 dollars in revenue for each premade banner sold.
- C. The store sold 14 custom banners yesterday.
- D. The store earns 14 dollars in revenue for each custom banner sold.

11

★ Mark for Review

ABC

The given function $h(t) = 100 - 4t$ models the number of hours until a gallon of milk held at a temperature of t , in degrees Celsius, goes sour. Which of the following is the best interpretation of the number 4 in this context?

- A. An increase of 1°C will make a gallon of milk go sour 4 hours faster.
- B. An increase of 1°C will make 4 gallons of milk go sour 1 hour faster.
- C. An increase of 4°C will make a gallon of milk go sour 1 hour faster.
- D. An increase of 4°C will make a gallon of milk go sour 4 hours faster.

12

★ Mark for Review

ABC

The given function $h(t) = 3.5 \left(\frac{t}{50} + 2 \right)$ models the height of a tree, in inches, t days after it is planted. According to the model, about how many inches does it grow each day?

A. 0.02

B. 0.07

C. 1.75

D. 3.5

13

★ Mark for Review

ABC

The given function $C(x) = 1.5 + 2.5x$ models the cost, in dollars, of mailing a shipment weighing x pounds. According to the model, an increase of 10 dollars in the mailing cost corresponds to an increase of how many pounds in the weight of the shipment?

A. 2

B. 2.5

C. 4

D. 5

14

★ Mark for Review

ABC

If $\frac{a}{b} = 4.5$ and $\frac{a}{kb} = 12.5$, what is the value of k ?

15

★ Mark for Review

ABC

In the given system of equations, k is a constant. If the system has no solution, what is the value of k ?

$$-5x = 8 - ky$$

$$18y + 5x = -10x + 21$$

16

★ Mark for Review

ABC

In the given system of equations, m is a constant. If the system has no solution, what is the value of m ?

$$\frac{4}{3} + \frac{5}{4}x = 28y + \frac{5}{8}x$$

$$my = \frac{1}{2}(5x - 8)$$

17

★ Mark for Review

ABC

In the given system of equations, b is a constant. If the system has no solution, what is the value of b ?

$$\frac{3}{10}y + \frac{b}{3}x = \frac{3}{4} - \frac{1}{5}y$$

$$\frac{3}{8}x + \frac{3}{5} = -\frac{1}{4}y + \frac{8}{5}$$

18

★ Mark for Review

ABC

In the xy -plane, the point $(18, y)$ is a solution to the given system of inequalities. Which of the following could be the value of y ?

$$x + y < 25$$

$$2x - 3y < 77$$

A. -14

B. -6

C. 7

D. 11

19

★ Mark for Review

ABC

Tina works no more than 30 hours at a nail salon each week. She can do a manicure in 20 minutes and a pedicure in 30 minutes. Each manicure earns her \$25 and each pedicure earns her \$40, and she must earn at least \$900 to cover her expenses. If m represents the number of manicures Tina does in a week and p represents the number of pedicures she does in a week, which system of inequalities describes all possible values of m and p for which Tina covers her expenses?

A. $3m + 2p \leq 30, 25m + 40p \geq 900$

B. $2m + 3p \leq 30, 25m + 40p \geq 900$

C. $\frac{m}{3} + \frac{p}{2} \leq 30, 25m + 40p \geq 900$

D. $\frac{m}{3} + \frac{p}{2} \geq 900, 25m + 40p \leq 30$

20

★ Mark for Review

ABC

During hibernation, adult male black bears lose 6 to 12 pounds in weight each week. An adult male black bear weighed 580 pounds after 5 weeks of hibernation. Which of the following inequalities gives all possible values for the black bear's weight w , in pounds, after 15 weeks of hibernation?

A. $400 \leq w \leq 490$

B. $460 \leq w \leq 520$

C. $460 \leq w \leq 580$

D. $610 \leq w \leq 640$

21

★ Mark for Review

ABC

A window repair specialist charges \$220 for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is \$400. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$

22

★ Mark for Review

ABC

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

$$\begin{aligned}8x + 7y &= 9 \\ 24x + 21y &= 27\end{aligned}$$

A. $\left(r, \frac{-8r + 9}{7}\right)$

B. $\left(\frac{-8r + 9}{7}, r\right)$

C. $\left(\frac{-8r + 63}{7}, \frac{9r + 189}{7}\right)$

D. $\left(\frac{r}{3} + 9, \frac{-r}{3} + 27\right)$

Answer Key — Unit I

No.	Answer		No.	Answer
1	-45		12	B
2	1.6		13	C
3	B		14	0.36
4	A		15	-6
5	A		16	112
6	A		17	9/4
7	C		18	B
8	A		19	C
9	C		20	B
10	B		21	A
11	A		22	A